

Perfectionism

Motivation and Workaholism

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COGS 164 - Neurobiology of Motivation



Background - Defining Terms

Workaholism = excessive, compulsive need to work; negatively impacts well-being.

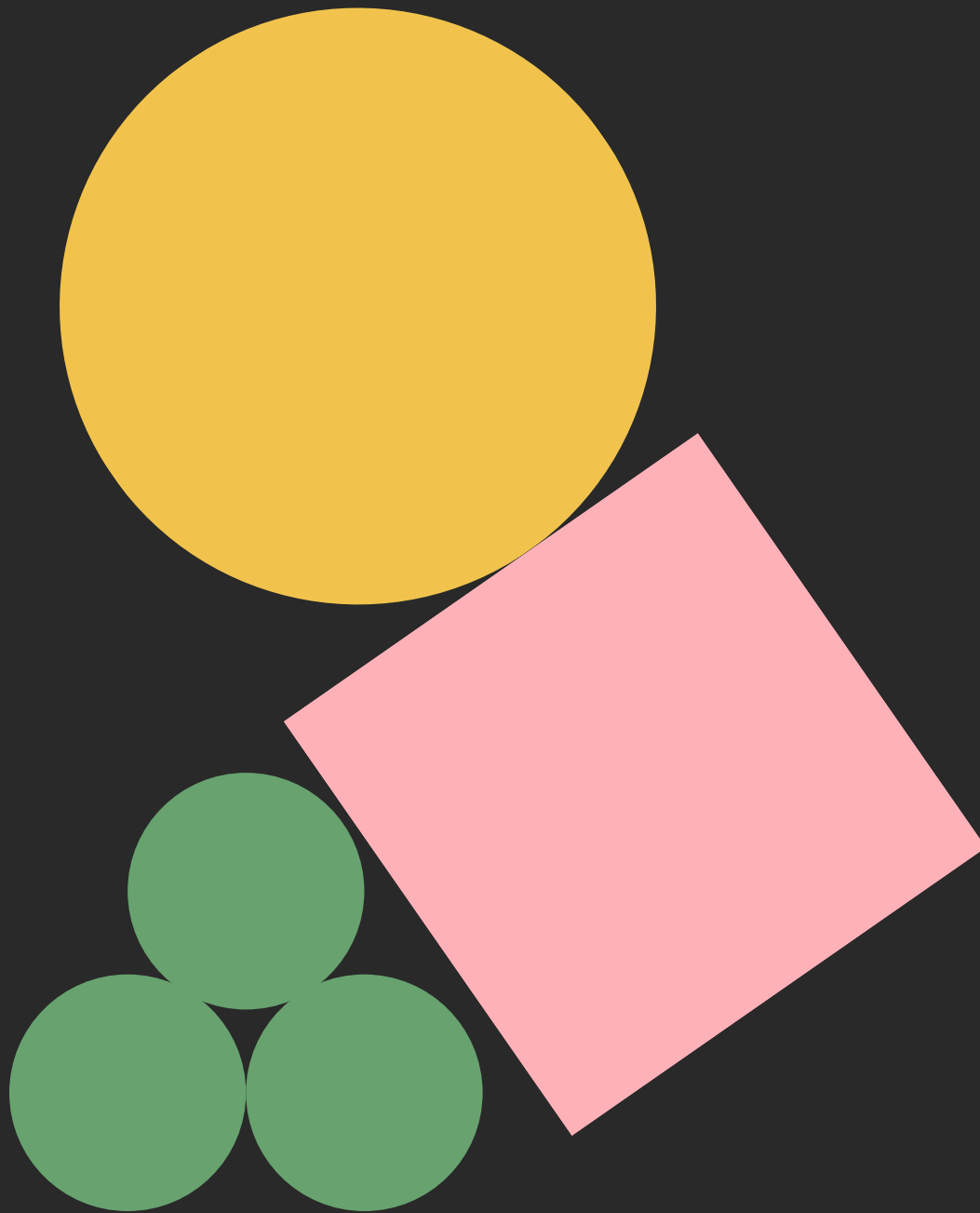
Perfectionism = setting high personal standards, fear of failure, self-criticism.

SOP Self-Oriented Perfectionism

- Internalized high standards
- Desire for self-improvement

SPP Socially-Prescribed Perfectionism

- External pressure
- Fear of judgment



Personal Standards



Evaluative Concerns

Neurobiology Background

Perfectionism is largely understood as a behavioral construct with a biological basis

Two biological aspects

- *Motivational regulation*
- *Emotional regulation*

Motivational regulation

- Dopaminergic reward system
 - Dopamine: largely associated to pleasure and reward
 - Origin: substantia nigra(SN), ventral tegmental area(VTA), and hypothalamus
 - Attention, aversion, reinforcement and motivation
 - Dopaminergic system performs various functions
 - Variation in dopaminergic modulatory pathways may differ a neurotypical from neurodivergent.

Dopamine in Motivational Control: Rewarding, Aversive, and Alerting(2010), The neuropsychological profile of work addiction.(2023), To err is (perfectly) human: behavioural and neural correlates of error processing and perfectionism(2017)

Emotional regulation

Primary characteristics of perfectionism

Anxiety and avoidance:

- Amygdala overactivation or dysfunction is correlated to generalized anxiety disorders, and when in conjunction with PFC(prefrontal cortex) , are involved with decision making and avoidant behavior.

Anxiety and self-criticism:

- Significant Variation in gray matter in emotion regulating regions(anterior cingulate/insular).
- These are regions believed to be key in error detection and processing leading to downstream emotional responses to outcome.
- Abnormality may be an explanation for frequent self-criticism and anxiety.

Neural systems underlying approach and avoidance in anxiety disorders(2010), Prefrontal Cortex and Impulsive Decision Making(2010), Perfectionism mediated the relationship between brain structure variation and negative emotion in a nonclinical sample(2016)

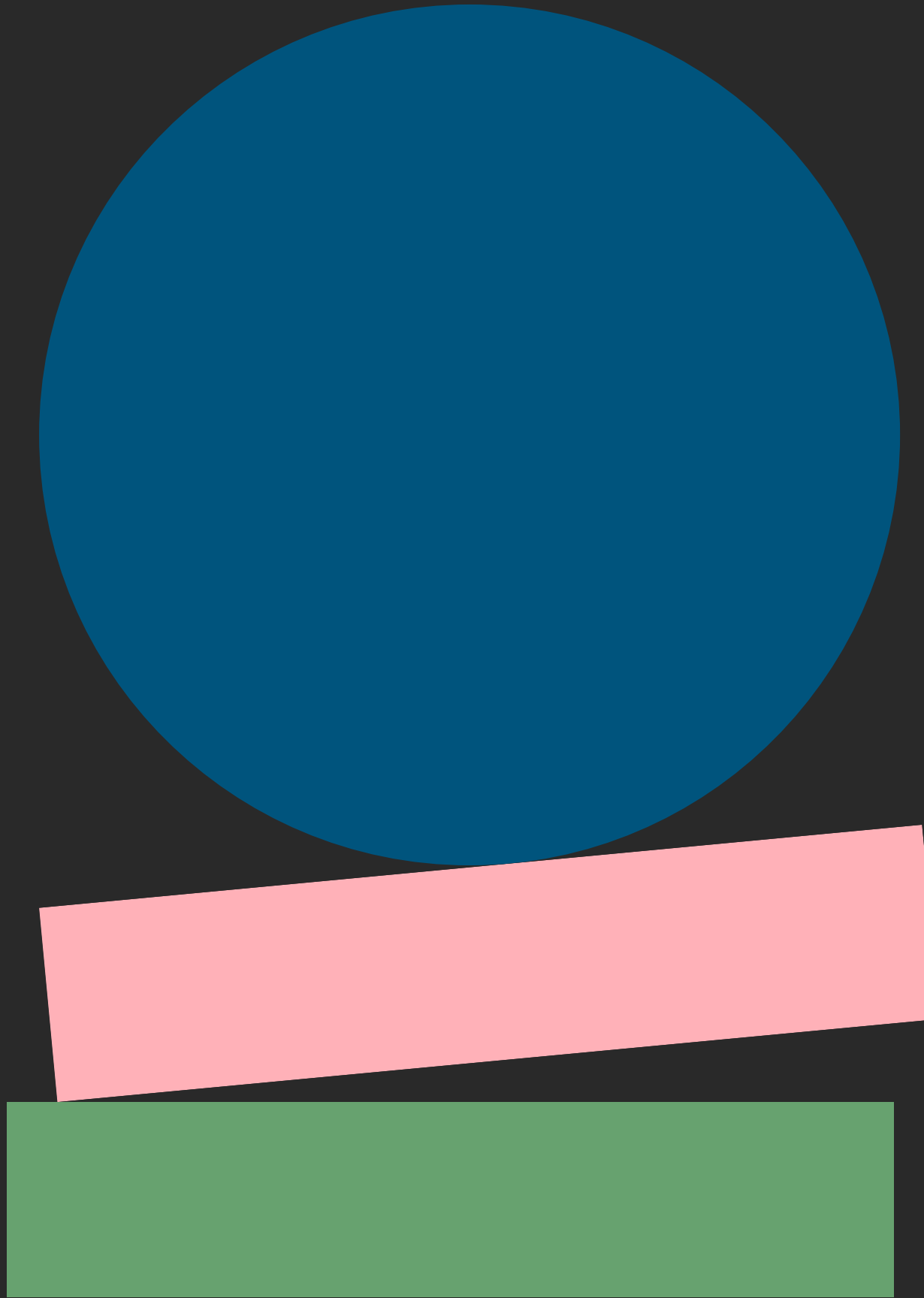
RESEARCH
QUESTION

Is Self-Oriented
Perfectionism (SOP) or
Socially Prescribed
Perfectionism (SPP) more
strongly associated with
workaholism, and do both
forms of perfectionism
contribute to unhealthy
work habits?

Hypothesis

We believe both SOP and SPP will be positively correlated with workaholism, but for different reasons:

- SOP:
 - intrinsically motivated
 - adaptive?
- SPP:
 - extrinsically motivated
 - maladaptive
 - --> stronger association with workaholism



METHODS

Multidimensional Perfectionism Scale (MPS)

- Identifying dominance for SOP and SPP
- 30 items following a Likert Scale: 1 (Strongly Disagree) to 7 (Strongly Agree)
- Scores range from 30 to 100

Work Addiction Risk Test (WART)

- Identifying the degree of work addiction
- 25 items on a Likert Scale: 1 (Never True) to 4 (Always True)
- Scores range from 25 to 100
- Higher Score = More Workaholism

Grading Methods

MPS

- Questions 1-15 = SOP
- Questions 16-30 = SPP
- Low perfectionism = 30 - 60
- Moderate Perfectionism = 61 - 90
- High Perfectionism = 91>

WART

- Low Risk = 25-56
- Medium Risk = 57-66
- High Risk = 67-100

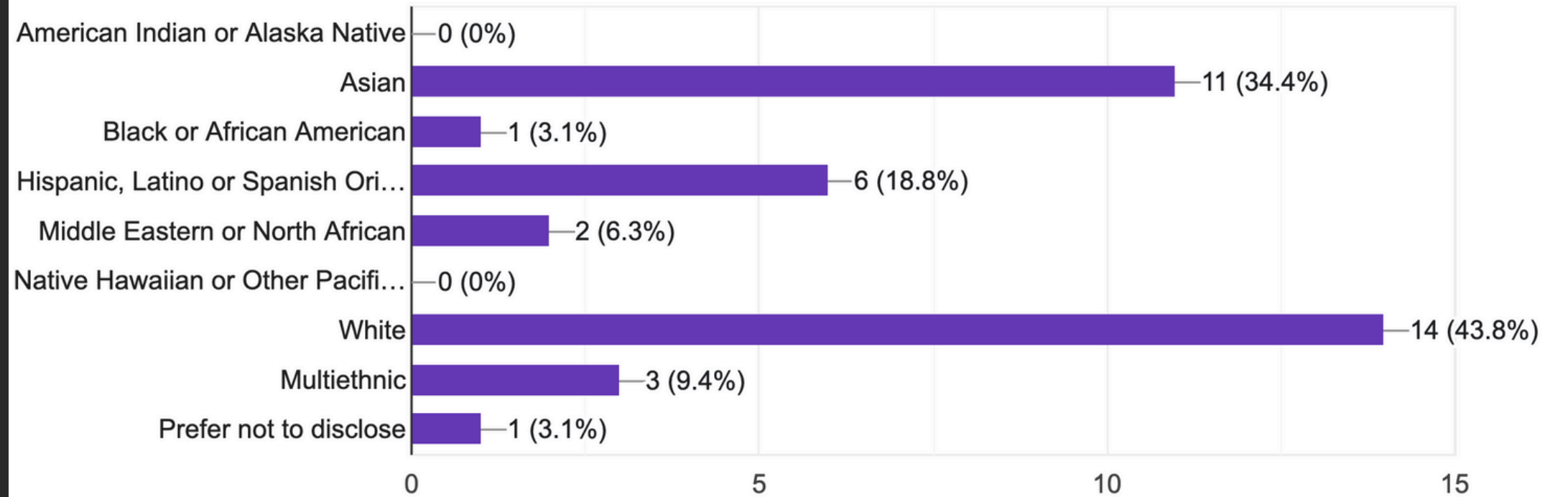
RESULTS

Demographics Of Survey Participants



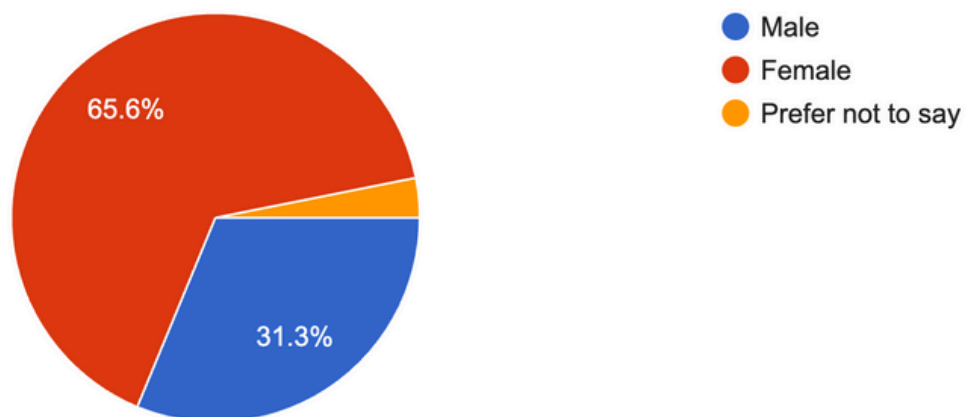
Which category describes you? Select all that apply.

32 responses



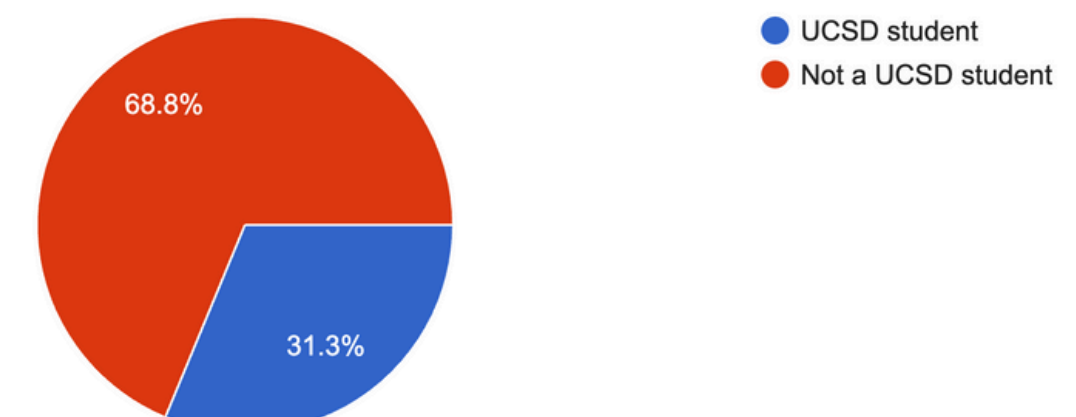
Sex

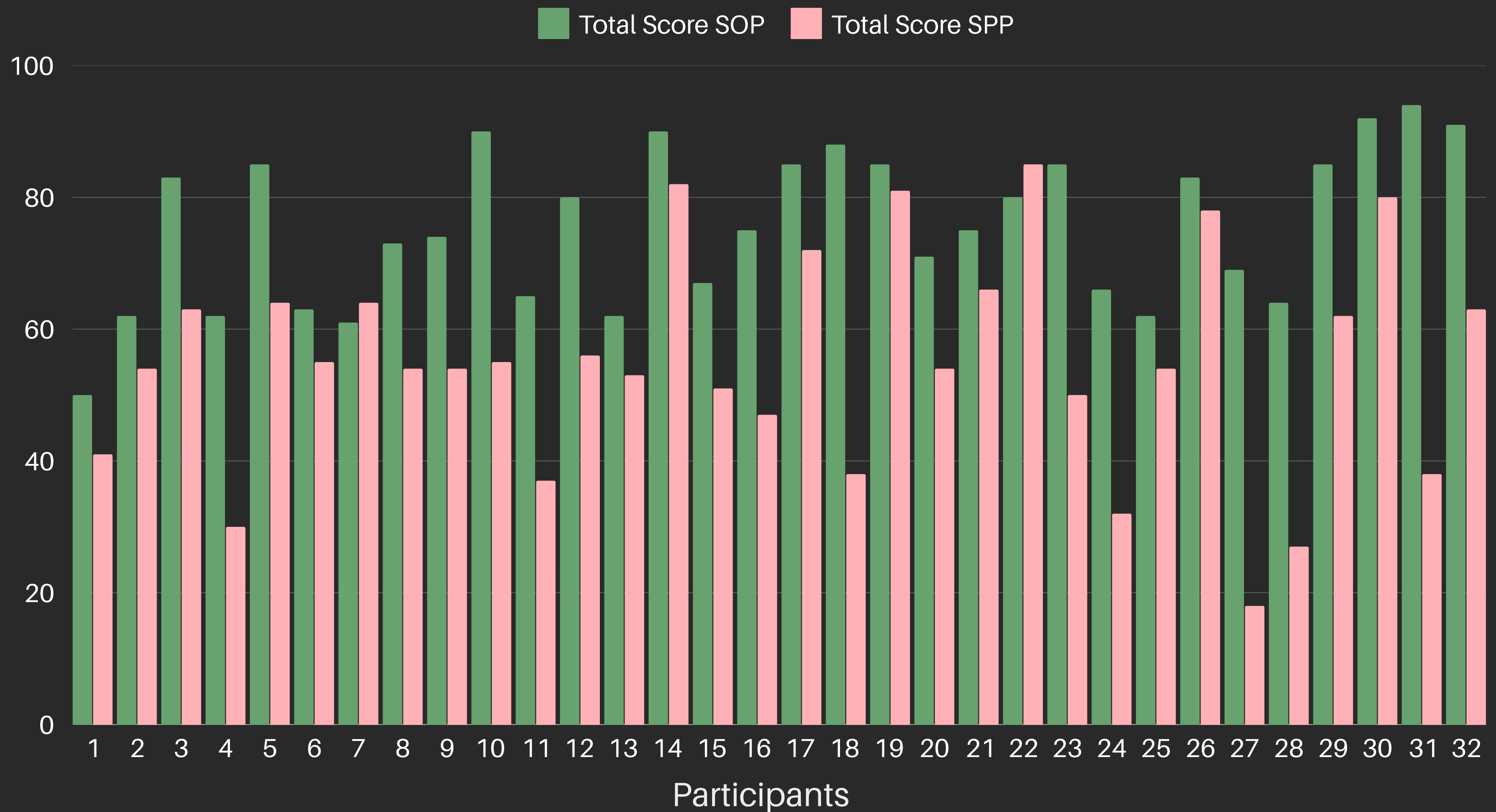
32 responses



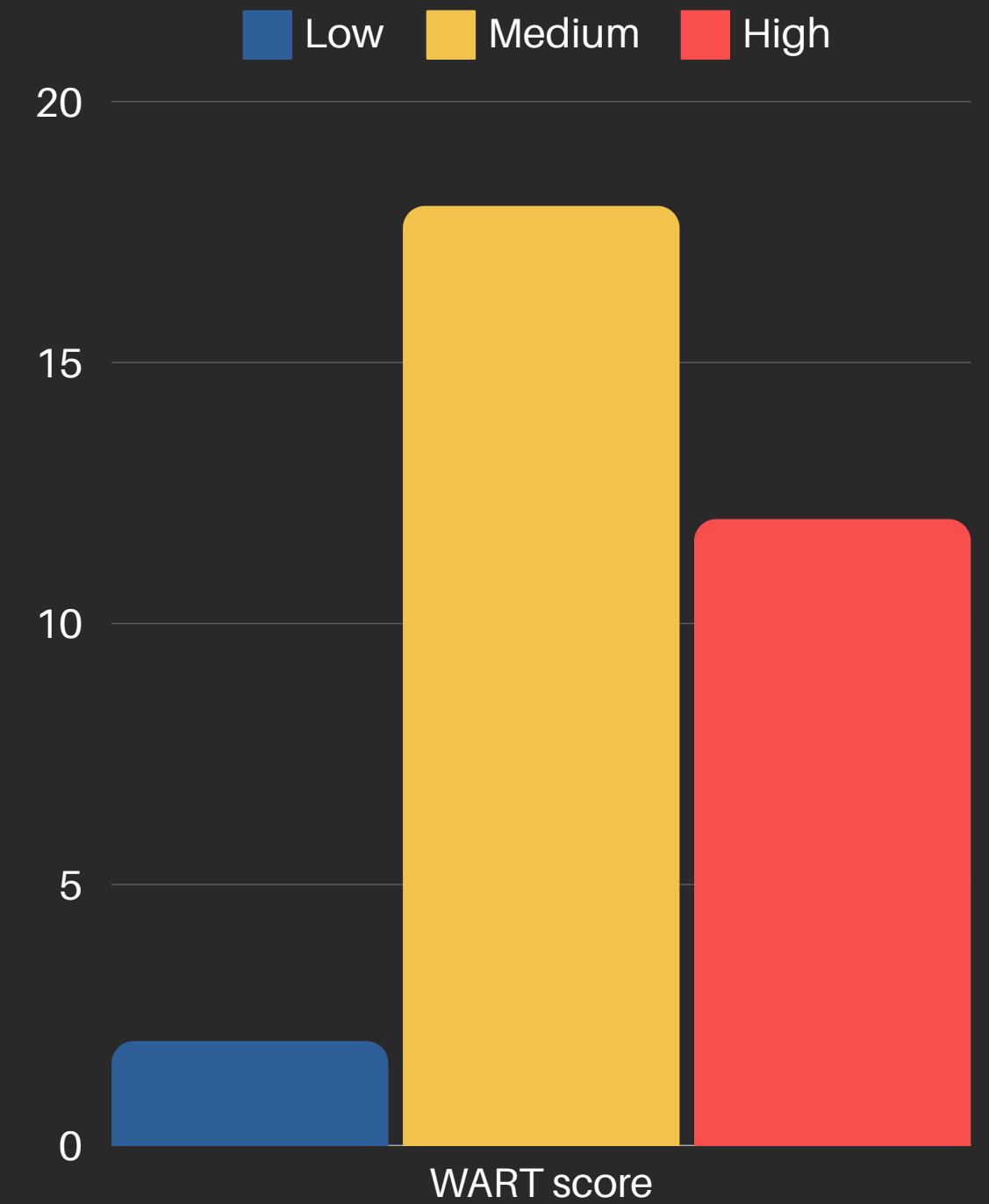
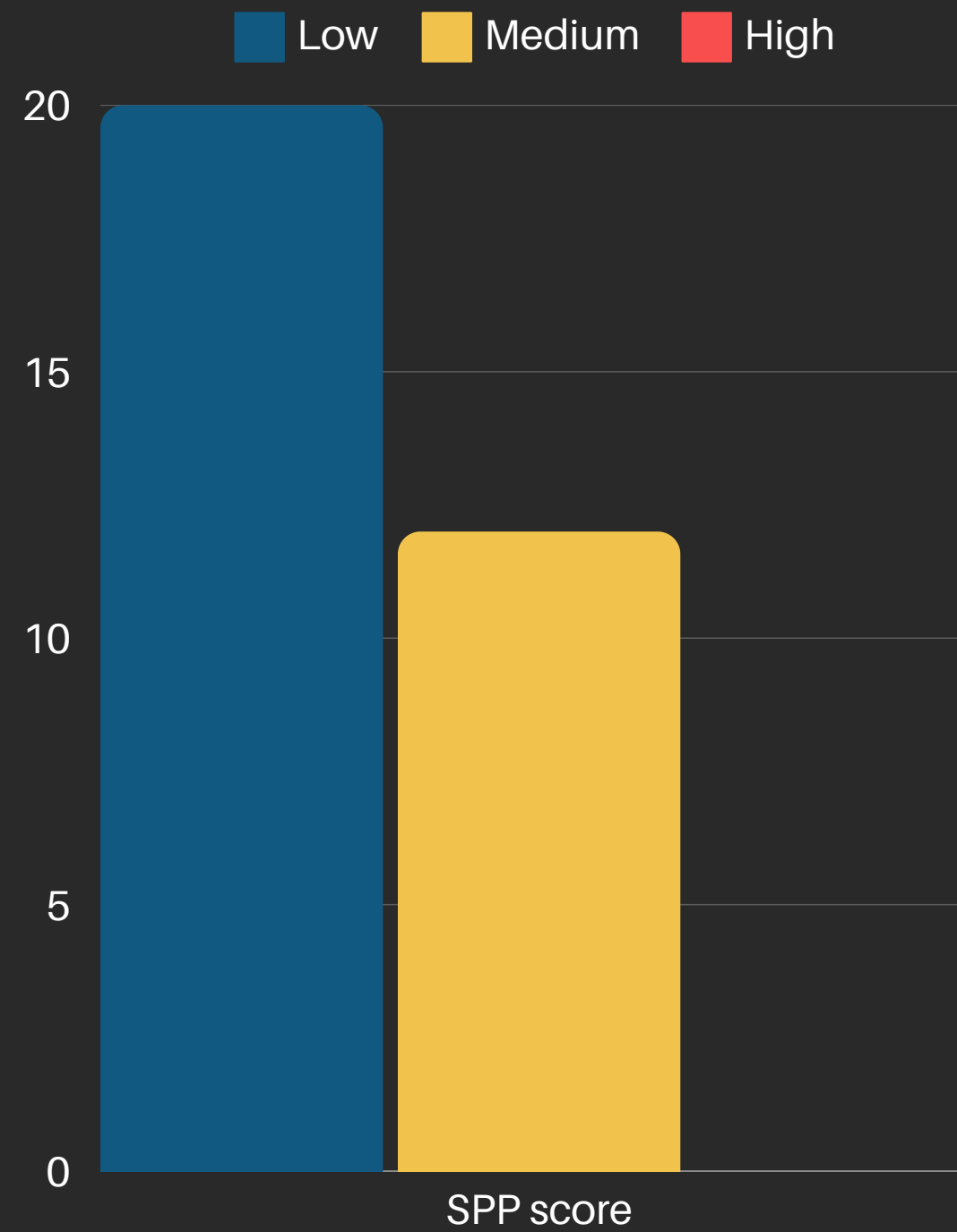
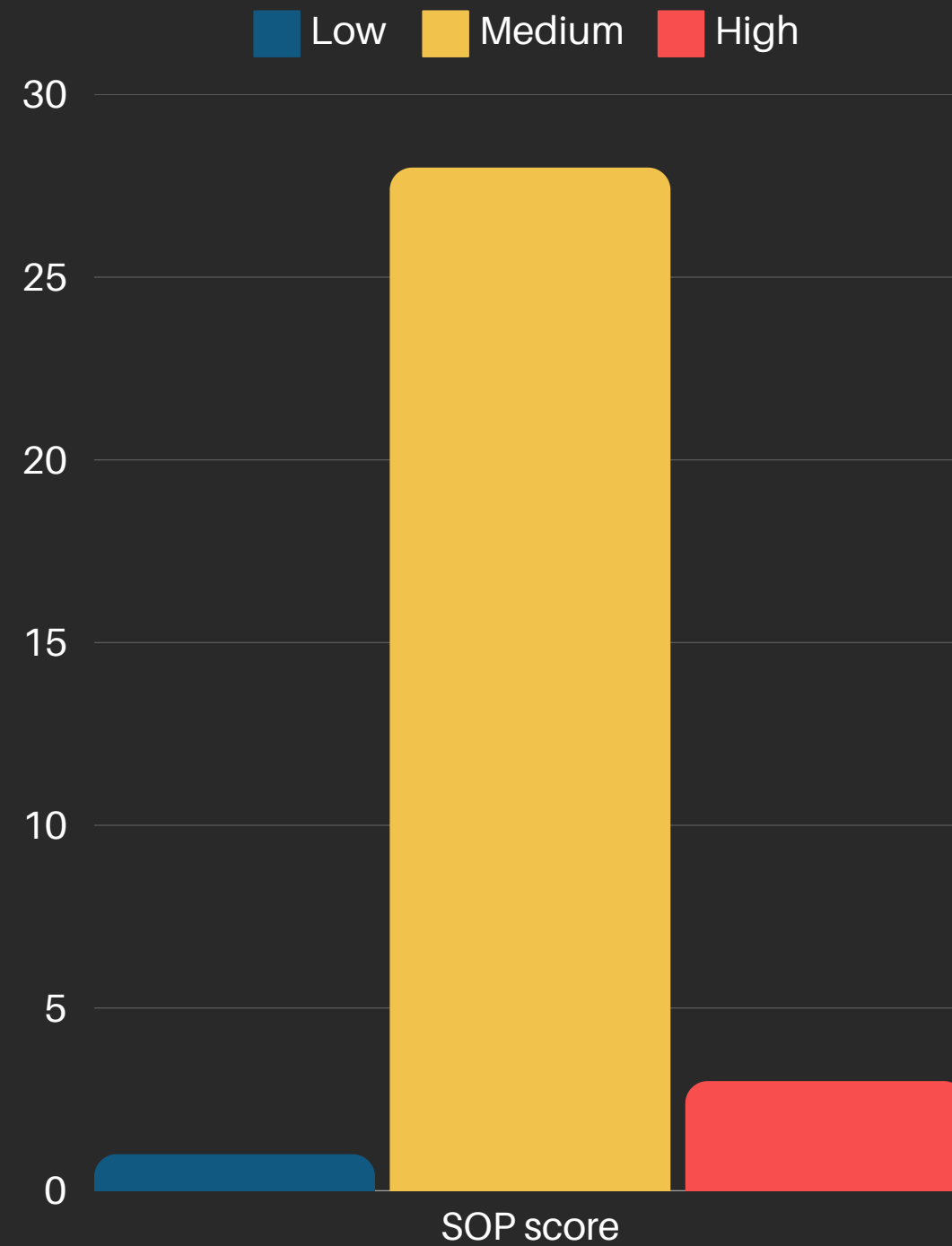
Are you a student?

32 responses

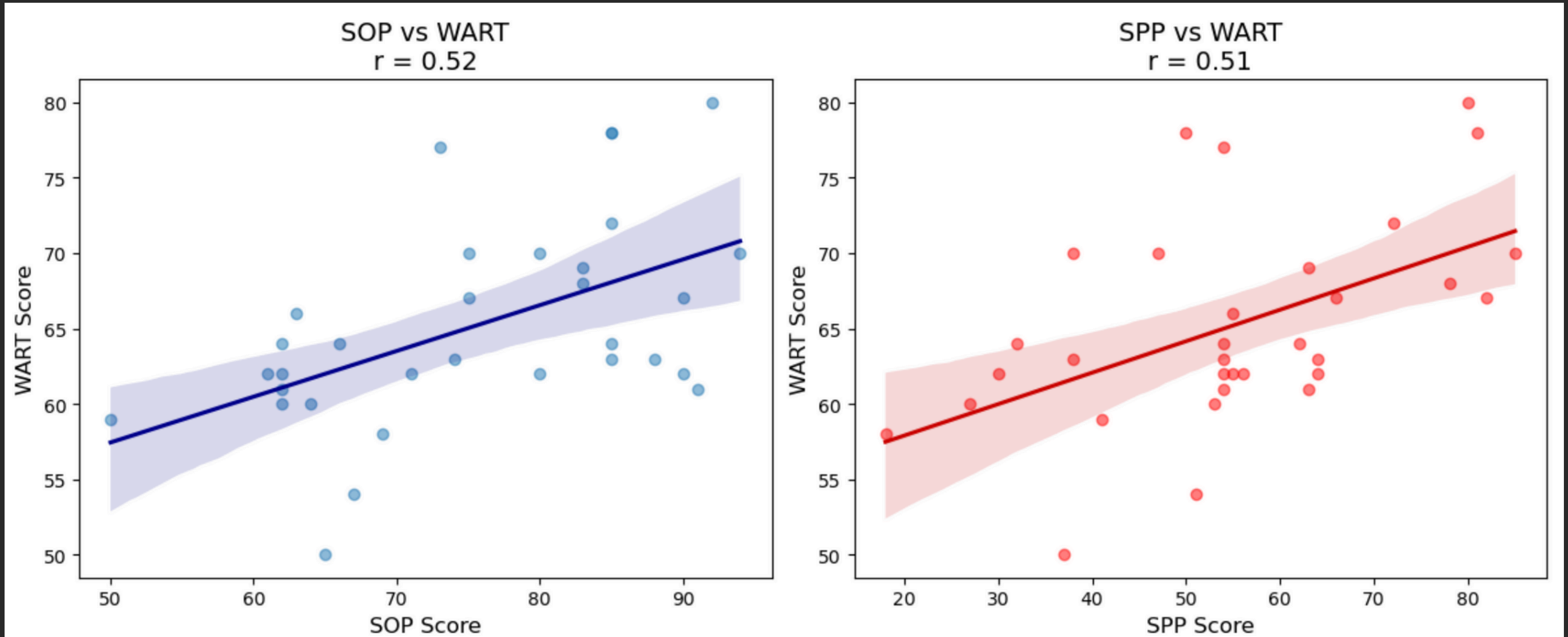




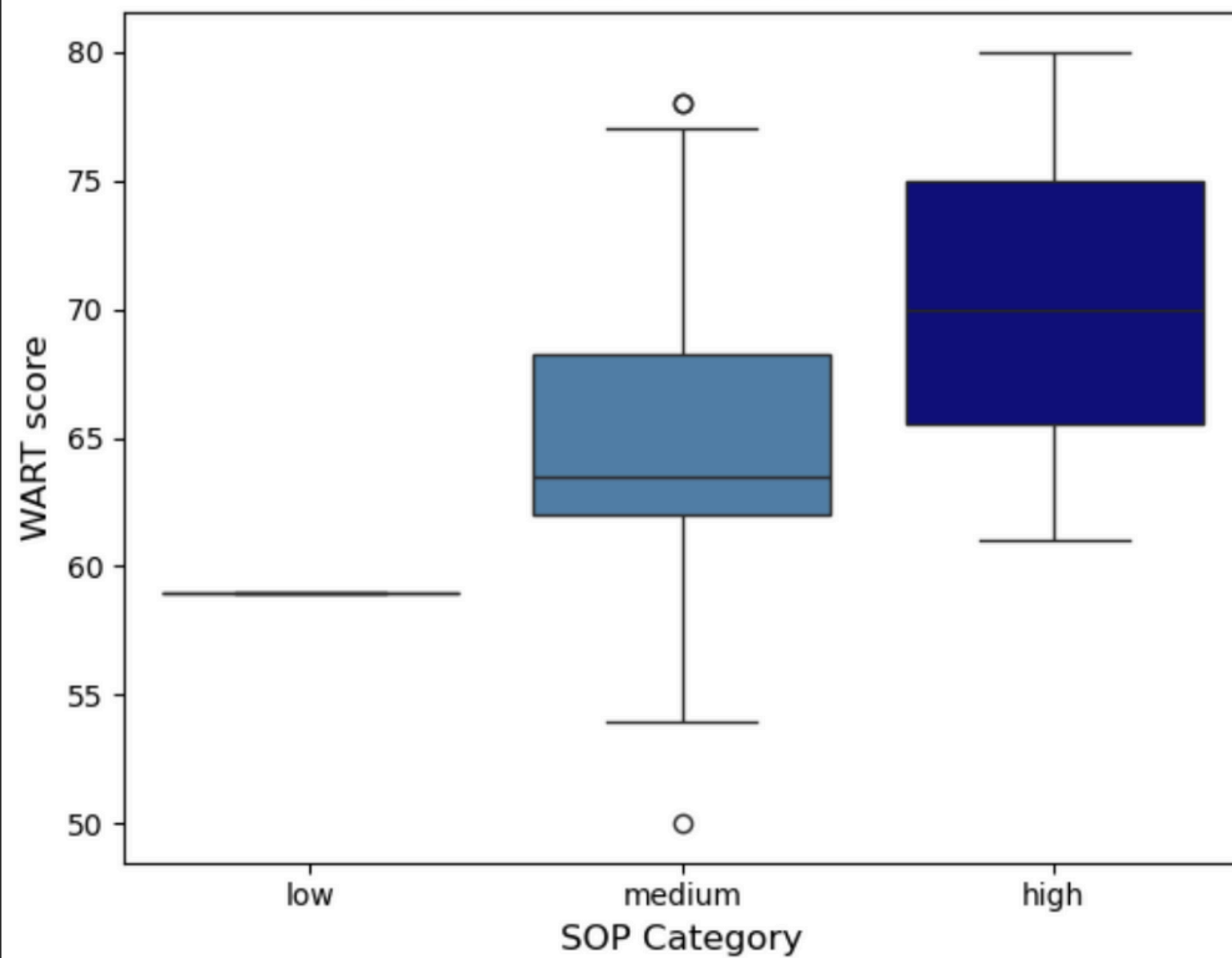
Categorizations of Results



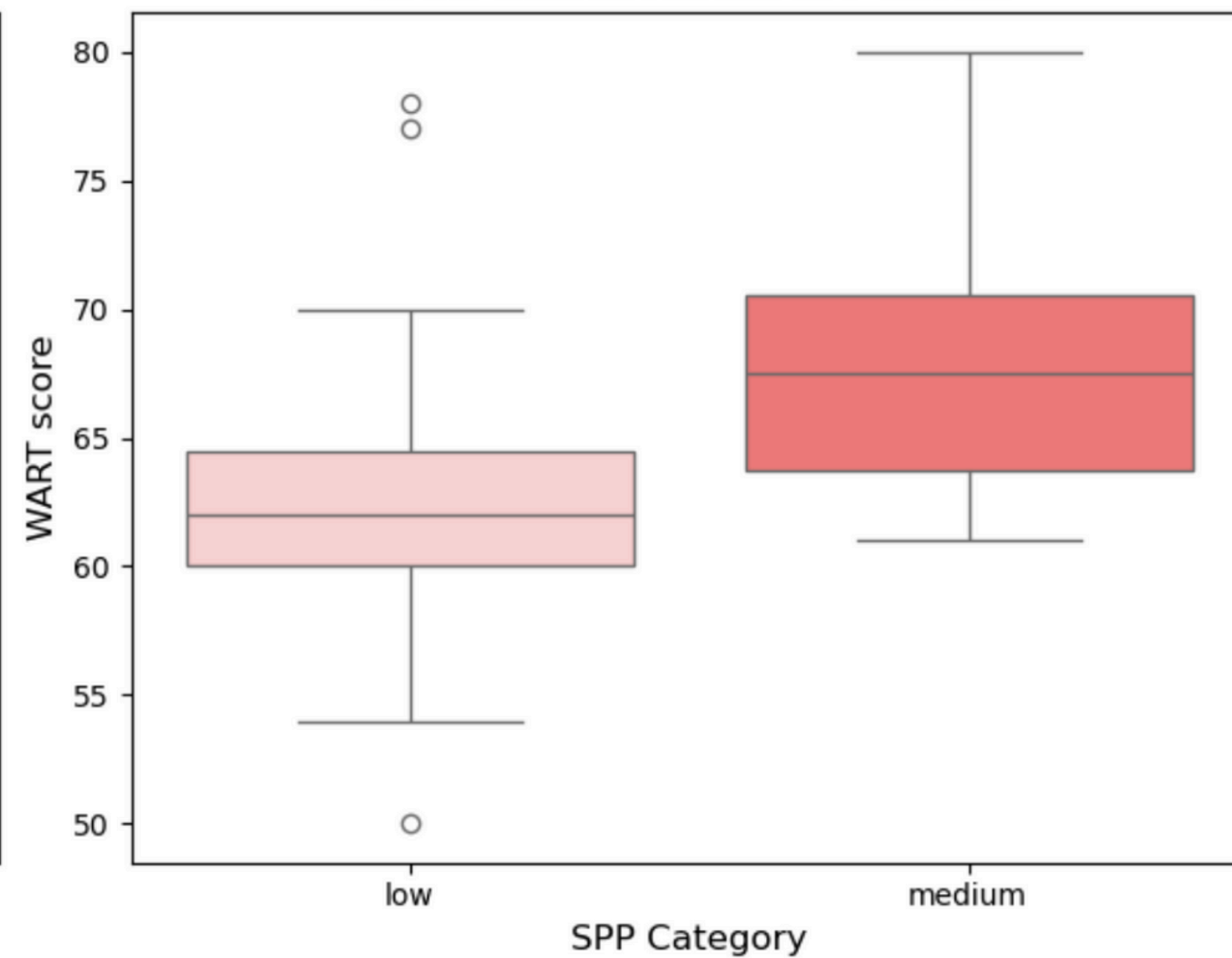
Both SOP & SPP are Positively Correlated with WART



WART Scores Across SOP Levels



WART Scores Across SPP Levels

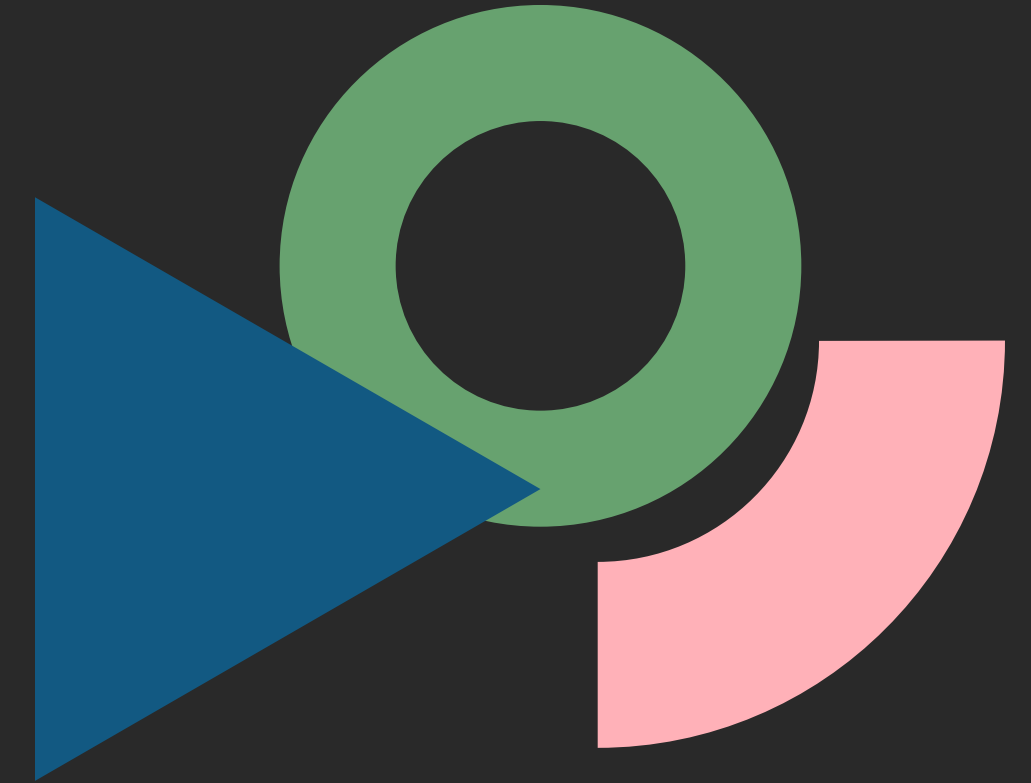


CONCLUSION

Perfectionism is positively linked to workaholism

*Both **SOP** & **SPP** show an equal association with workaholism*

Limitations



Self Report
Bias -
participant
consistency
in answering
questions

Could have
overlap in
SOP vs. SPP
groups

Not using the
complete MPS
and WART -
chose fewer
questions

Quantitative
data without
giving
participants
qualitative
explanations

Reasons
outside of
compulsion -
passion or not
letting work
stack up



Team Contributions

Anna Lin

- background research on work motivation, perfectionism and workaholism
- data visualizations in synthesizing survey results

Roxana Chen

- ideation & literature review
- experimental design & data analysis

Stephanie Castillo

- Methods
- Created MPS/WART survey
- Graded MPS/WART

Angelica Petrov

- additional research on perfectionism vs workaholism
- grading responses

Luke Yang

- Neurobiology basis/background research/Literature review
- Optimization and Error Detection with other resources

Everyone

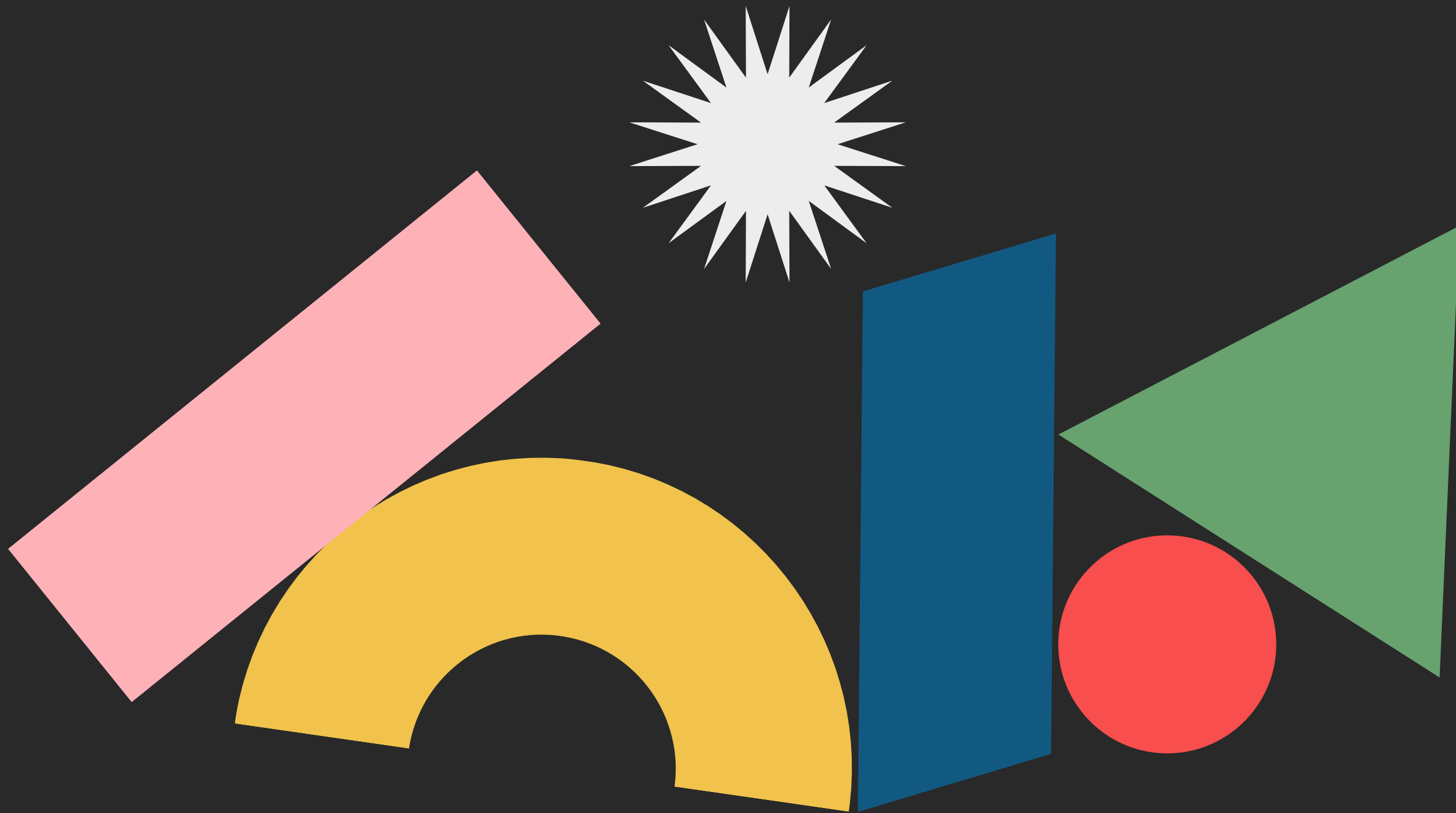
- sample search - 5 survey participants recruited each

Resources

- Psychometric assessment of workaholism measures - WART questions
- Multidimensional Perfectionism Scale
- Perfectionism and Workaholism in Employees
- Dopamine in motivational control: rewarding, aversive, and alerting
- Neural systems underlying approach and avoidance in anxiety disorders
- Perfectionism mediated the relationship between brain structure variation and negative emotion in a nonclinical sample
- To err is (perfectly) human: behavioural and neural correlates of error processing and perfectionism
- The neuropsychological profile of work addiction
- Prefrontal Cortex and Impulsive Decision Making
- Perfectionism: Intrinsic vs. Extrinsic Motivation
- The 2x2 Model of Perfectionism
- Perfectionism in Employees: Work Engagement, Workaholism, and Burnout
- Workaholism: Being Driven to Work Excessively Hard



Thank you!



For Grading

Link to Project Pitch slides:

[https://www.canva.com/design/DAGfDjKu1vM/ArT27hefFv2r6sT-Nmz79A/edit?
utm_content=DAGfDjKu1vM&utm_campaign=designshare&utm_m
edium=link2&utm_source=sharebutton](https://www.canva.com/design/DAGfDjKu1vM/ArT27hefFv2r6sT-Nmz79A/edit?utm_content=DAGfDjKu1vM&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton)